

N

- name errors, 17–18
- name_function.py*, 211–217
- name.py*, 20
- names.py*, 211–212
- nesting. *See* dictionaries: nesting; lists:
 - for loops
- newline (`\n`), 21–22
- `next()` function, 330–331
- None, 98, 140
- number_reader.py*, 202
- numbers, 26–28
 - arithmetic, 26
 - constants, 28
 - exponents, 26
 - floats, 26–27
 - formatting, 291–292
 - integers, 26
 - mixing integers and floats, 27–28
 - order of operations, 26
 - `round()` function, 291–292
 - underscores in, 28
- number_writer.py*, 201

O

- object-oriented programming
 - (OOP), 157. *See also* classes
- or keyword, 76. *See also* if statements

P

- pandas, 320
- parameters, 131
- parrot.py*, 114, 118–121
- pass statement, 198–199
- paths. *See* files: file paths
- PEP 8, 68–69
- person.py*, 139–140
- pets.py*, 125, 132–136
- pip, 210–211
 - installing Django, 374–376
 - installing Matplotlib, 302
 - installing Plotly, 320
 - installing Pygame, 228
 - installing pytest, 211
 - installing Requests, 357
 - Linux, installing pip, 465–466
 - updating, 210

- pi_string.py*, 187–189
- pizza.py*, 146–148
- Platform.sh. *See* Django: deployment
- players.py*, 61–62
- Plotly, 302, 319. *See also* mapping
 - earthquakes; rolling dice
- chart types, 322
- customizing plots, 323, 325–326, 364
- documentation, 368
- `fig.show()` method, 322
- `fig.write_html()` method, 327
- formatting plots
 - axis labels, 323
 - color scales, 349–350
 - hover text, 350–351, 365–366
 - links in charts, 366–367
 - marker colors, 349–350, 367
 - tick marks, 325–326
 - titles, 323
 - tooltips, 365–366
 - `update_layout()` method,
 - 325–326, 364
 - `update_traces()` method, 367
- gallery, 320
- histograms, 322
- installing, 320
- `plotly.express` module, 322,
 - 347, 368
- px alias, 322
- `px.bar()` function, 322–323, 363–367
- saving figures, 327
- `scatter_geo()` function, 347–352
- `pop()` method, 39–40
- positional arguments, 131–133. *See also*
 - functions: arguments
- POST requests, 406. *See also*
 - Django: forms
- printing_models.py*, 143–145
- Project Gutenberg, 196–197
- prompts, 114–115
- `.py` file extension, 15–16
- Pygame. *See also* Alien Invasion
 - background colors, 231–232
 - `clock.tick()` method, 230–231
 - collisions, 266–267, 270–271,
 - 289–290
 - creating an empty window, 229–230
 - cursor, hiding, 282–283